

IEEE 830 • ISO/ IEC 29148.

From culture to structure — what a Software Requirements Specification actually obliges you to write, and why ISO 29148 superseded IEEE 830.

LECTURE · 90 MIN

Standards & Templates

90-min lab on building your own SRS + SDD templates.

Today's Agenda

90-minute lecture, six blocks, one exercise to M2

00–10

Entry quiz & W1 homework check

Five-minute retrieval on Week 1, three-question M1 review.

10–35

IEEE 830 vs. ISO/IEC/IEEE 29148

Origin, structure, traceability, when each fits your project.

35–60

SRS & SDD structure in detail

Mandatory sections, quality attributes of a good requirement, traceability matrix.

60–85

Template analysis — good vs. bad SRS excerpts

Working session: rewrite three weak sentences into testable requirements.

85–90

Exercise handover — M2 deadline

Build SRS + SDD template, draft five IEEE-style requirements, attach traceability rows.

Recap: Week 01

Five ungraded retrieval questions — written, no peer chatter.

FIVE QUESTIONS · 5 MIN

1. Name the four documentation types introduced in W1.
2. Which anti-pattern explains “Only Ali can deploy”?
3. What is the target reader of a SRS?
4. Why does the “Good culture” column matter more than tools?
5. Give one example of “defensible omission” from your own project idea.

Why a written retrieval? Retrieval practice at the start of a session roughly doubles retention at week 12 (Roediger & Karpicke, 2006).

IEEE 830 vs. ISO/IEC/IEEE 29148

Why two standards exist, where they overlap, and which one your contract probably names.

DIMENSION	IEEE 830-1998	ISO/IEC/IEEE 29148:2018
Origin & lineage	1998 IEEE Computer Society — the SRS template most curricula still teach first.	Joint ISO / IEC / IEEE — supersedes 830; extends to requirements process and stakeholder needs.
Mandatory sections	Five top-level (Intro, Overall, Specific, Appendices) — mixed content under Specific.	Eight documents: StRS, SyRS, SRS, SAD, SDD, etc. — cleaner separation by concern.
Requirements quality	Implicit — recommended rather than enforced.	Eight named quality attributes — Correct, Unambiguous, Complete, Consistent, Ranked, Verifiable, Modifiable, Traceable.
Safety / regulated	Often named in legacy aerospace / medical contracts.	Implied baseline in any new regulated procurement in the EU / Mongolia / ISO markets.
Use in this course	Teaching reference — primary template students map first.	Compliance target — the eight quality attributes drive M3 grading.

DECISION RULE OF THUMB

Use IEEE 830 as your structural template; use the ISO 29148 quality attributes as your acceptance criteria. The two are not competitors — they are layers.

SRS Sections & Quality Attributes

ISO 29148 names eight attributes a requirement must satisfy to count.

REQUIRED SRS SECTIONS (IEEE 830 MAPPING)

- 1 · Scope, product, audience, constraints
- 2 · Referenced documents & standards
- 3 · Specific requirements — functional, performance, design constraints
- 4 · External interface requirements (user, hardware, software, communication)
- 5 · Non-functional requirements (performance, security, reliability, maintainability)
- 6 · Other requirements — legal, regulatory, audit
- 7 · Appendices — glossary, traceability matrix, assumptions

8 ISO 29148 QUALITY ATTRIBUTES

Correct · Unambiguous · Complete · Consistent

Ranked for priority · Verifiable · Modifiable · Traceable

If you cannot write a black-box test for your requirement, it is decoration. The gradable test of “Verifiable” is: in one sentence, name the input, the action, the observable output, and the pass / fail boundary.

Template Analysis — Good vs. Bad

Four weak requirements from real student projects, four repaired readings.

BAD · AMBIGUOUS

"The system shall be fast."

REPAIR · VERIFIABLE

"The search endpoint shall return results within 300 ms for 5×10^4 catalogue entries on the reference VM."

BAD · NOT TESTABLE

"The user interface shall be user-friendly."

REPAIR · RANKED & OBSERVABLE

"Nielsen's ten heuristics: a heuristic review with five target users shall reach a SUS score ≥ 75 ."

BAD · DESIGN MASQUERADING

"The system shall use PostgreSQL."

REPAIR · SPLIT SRS / SDD

SRS: "Persistent state must satisfy ACID on a single transaction." SDD: "PostgreSQL 16 is the chosen RDBMS."

BAD · MISSING BOUNDARY

"The system shall handle many users."

REPAIR · MEASURABLE CEILING

"The API must sustain 200 concurrent requests at $p95 < 500$ ms with $< 1\%$ error rate."

CONCEPT TEST

Vote individually · 30-second peer defence · re-vote · reveal

QUESTION 1 / 1

Your team needs an SRS for a regulated procurement. The contract annex names only “IEEE 830-1998”. Which standard’s quality attributes should you still apply to your content?

A

Only IEEE 830. The contract is the contract.

B

IEEE 830 structure – but ISO 29148 attributes as the acceptance criteria.

C

Re-write the contract annex – skip IEEE 830 entirely.

D

Use both: ISO 29148’s eight documents replace IEEE 830 inside the same annex.

REVEAL ANSWER — AFTER VOTES

B is correct.

Structure (the table of contents) follows the contract; quality criteria follow ISO 29148 – they are not in conflict. Doing only A makes the document legally compliant but useless; doing C risks contractual breach; doing D is a clever over-step that triggers a procurement re-bid.

Take-aways & Next Week

You have the structure. Now you write the document.

THREE TAKE-AWAYS FROM WEEK 02

1. **Two standards stack — IEEE 830 is your table of contents; ISO 29148's eight attributes are your quality bar.**
2. **A requirement is what you can black-box test — “user-friendly” is not one.**
3. **Keep design out of the SRS — the SDD is where it lives.**

M2 DELIVERABLE · DUE FRIDAY

Map IEEE 830 outline to your project, mark gaps, draft five IEEE-style requirements with traceability rows. Build SRS and SDD templates in your workspace.

NEXT WEEK · WEEK 03

Writing the SRS in full.

By next Friday (M3, the first Pflicht-Meilenstein): fifteen functional plus five non-functional requirements, prioritised, in your workspace. We also introduce the KI-Workshop on how to use GPT-4 to generate SRS candidates and the verification protocol to check them.